

Title: Integration of endogenous and environmental signals in the shoot apical meristem by redox signaling

Abstract:

Developmental plasticity is a defining feature of plants, enabling them to dynamically adjust their shape and physiology in response to changing environmental conditions. This plasticity is largely made possible by the post-embryonic activity of meristems, stem cell-containing tissues responsible for generating all plant organs. The shoot apical meristem (SAM), in particular, gives rise to aerial organs such as leaves, stems, and flowers through tightly regulated processes of stem cell maintenance, proliferation, and differentiation. Among environmental cues, light plays a central role in developmental regulation, as optimizing light capture is directly linked to photosynthetic efficiency and growth. In the context of climate change—characterized by rising temperatures, altered rainfall patterns, and increased planting densities—understanding shade responses is essential for enhancing crop performance and resilience. While shade avoidance has been well-characterized in traits such as hypocotyl elongation and leaf positioning, its impact on SAM function remains largely unexplored. This project focuses on how the SAM integrates light cues, particularly during shade avoidance—a competitive response to neighboring vegetation. In particular, we have identified redox signaling as a promising point of interaction between endogenous and environmental cues in the regulation of SAM activity during vegetative development in *Arabidopsis*. Thus, the specific aims of the project will be to 1- characterize how shade affects SAM function by assessing changes in stem cell activity, proliferation, and differentiation, and mapping the spatial and temporal dynamics of redox signals using developmental markers, confocal microscopy, histochemistry, and genetic tools; 2- determine the functional significance of redox signaling in SAM regulation under shade conditions through pharmacological and genetic approaches; 3- investigate the impact of redox signaling on post-translational modifications (PTMs) on a global level using variants of biotin-switch assays and in particular evaluating the role of specific candidates, which have been shown to have a role in other developmental processes. This Research Partnership Grant will be instrumental to start a new collaboration between two newly established research groups led by young female scientists, with complementary skills and interests. The Swiss research lab will contribute with a vast experience on plant developmental responses to light signals, while the Latin American counterpart will contribute with all the required theoretical and technical skills to work on redox signaling in *Arabidopsis*. This seed funding will allow us to set the basis of the role of redox signals on light responses in the SAM, leading to a joint first publication within one year, and importantly providing groundwork for future long-term projects.

Relevance and justification

This project builds a strategic international collaboration between a Swiss group with expertise in light signaling and a Latin American team specializing in redox biology—expertise not currently available in Switzerland. The partnership combines complementary technical skills in imaging, developmental biology, photobiology, and redox-based biochemical and proteomic methods to address a key biological question from a multidisciplinary perspective. It will also foster early-career training through internships and joint supervision. The requested funding is essential to launch this collaboration, generate preliminary data, and lay the foundation for future large-scale competitive grant applications.

State of the art, hypotheses and aims.

One key feature of plant biology is their developmental plasticity, which allows them to adapt their growth, development and physiology to the prevailing environmental conditions. This is possible because most of a plant's development occurs post-embryonically due to the activity of meristems, groups of stem cells that through proliferation and differentiation give rise to all plant organs. Aerial organs, in particular, originate in the shoot apical meristem (SAM)¹.

Among environmental cues, light is a major regulator of plant development, providing information about day-length, position, and presence of competitors, among others. Predictably, light plays a central role in regulating SAM activity during the transition from dark to light in de-etiolating seedlings²⁻⁴. This involves energy status sensing and photoreceptor activity, including phytochromes³. Phytochromes are red and far-red light receptors which upon light perception regulate the transcriptome through interaction with transcription factors (e.g. PIFs) or transcriptional regulators such as the E3 ligase COP1⁵. Notably, COP1 has been identified as a key player in mediating light-induced activation of the SAM³.

Given that photosynthetic tissues absorb red and blue light but transmit and reflect other wavelengths, the light environment surrounding a plant presents a low red to far-red ratio (R:FR)⁶. This low R:FR condition inhibits phytochrome activity, triggering shade avoidance responses, including: stem growth, leaf development and flowering transitions⁷. Shade responses can be experimentally induced by far-red light treatments that lower the R:FR without affecting photosynthetically active radiation (PAR), thereby minimizing the effect of light treatments on the energy status^{5,8-11}. Moreover, it is an ecologically relevant signal as plants respond to changes in the R:FR caused by FR reflection from other plants.

Importantly, FR treatment has been shown to affect the transcriptome of the shoot apex in seedlings¹². Moreover, upon a FR treatment, a rapid and transient arrest in cell proliferation in the leaf primordia has been identified, which could account for the reduced blade area found in shade-avoiding Arabidopsis plants¹³⁻¹⁵. This reduction in cell proliferation has been linked to a reduction in cytokinin activity¹³ as well as to the regulation of key cell cycle genes by PIF7 and AN3^{14,15}. Interestingly, in tomato, shade induces contrasting effects by promoting leaf blade expansion, possibly due to species-specific differences in SAM responses during early development¹⁶. Shade also transiently modulate SAM size¹⁶. Moreover, it has been shown that phytochromes and shade regulate stomata density early in primordia development, both in tomato and in Arabidopsis^{17,16}.

Interestingly, these responses to light rely on light perception outside the SAM, but where they are perceived, and the signals linking the site of light perception with the site of response remain unknown^{1,3,16,18}. Moreover, whether shade signals can affect other aspects of the SAM activity, including stem cell homeostasis has not been studied.

Emerging evidence highlights the role of reactive redox species including hydrogen peroxide (H_2O_2), superoxide anion (O_2^-), nitric oxide (NO), and hydrogen sulfide (H_2S) as key signaling molecules in plant development and environmental responses. These redox signals possess unique biochemical properties: they are small, diffusible, and highly but specifically reactive; they act in part by modifying protein function through post-translational modifications (PTMs); and are strictly regulated in response to environmental cues¹⁹⁻²³. Notably, it has been recently shown that shade rely on redox signaling to regulate seedling development, with site-specific effects on cotyledons and hypocotyls²⁴. In particular, COP1 and some of its targets have been identified as target of redox PTMs, linking redox regulation directly to central components of light signaling pathways^{25,26}. Within the SAM, redox signaling also appears to be critical for balancing cell proliferation and differentiation^{1,25,27,28}. A tight spatial compartmentalization within the SAM has been shown: O_2^- accumulates in the central stem cell zone to promote WUSCHEL expression and stemness, whereas H_2O_2 and NO are enriched in the peripheral zone, where they promote differentiation. Disruption of this redox gradient significantly reduces SAM size and delays early leaf development^{27,28,29}. In addition, H_2S biosynthesis genes are both regulated by shade and between Arabidopsis SAM clusters in single cell RNAseq studies^{8,30}.

Taken together, these findings reveal a compelling intersection between light and redox signaling in the control of plant development. However, the role of redox signaling as a potential mediator that links light perception outside the SAM to its developmental regulation remains largely unexplored. Building on this, **we hypothesize that reactive redox species act as key intermediates in the transduction of light quality signals from peripheral perception sites to the SAM, thereby modulating its activity and developmental outputs.**

Project Objectives

1: Characterize the impact of shade on SAM function in *Arabidopsis* seedlings

1.1 Investigate the effects of shade on stem cell homeostasis, proliferation, and differentiation within the SAM.

1.2 Analyze the spatial and temporal dynamics of redox signals within the SAM in response to shade.

2: Elucidate the functional role of redox signaling in the environmental control of SAM regulation

3: Determine whether redox signals are transduced through cysteine-based PTMs

3.1 Identify global changes in protein redox-mediated PTM under shade conditions.

3.2 Investigate the role of COP1 as a redox-sensitive target involved in regulating SAM activity in response to light quality cues.

Research plan, methods and timeframe.

In the first phase of the project, we will explore the effect of shade signals on the shoot apex development and redox signaling. In order to change photoreceptor activity without changing the energy status of the plant, we will keep the photosynthetically active radiation constant, and reduce phytochrome activity by adding far-red light (FR, 735nm) with wavelength-specific LEDs, which is a well-established procedure in the field^{10,12,16}.

To assess how shade affects SAM activity in *Arabidopsis thaliana*, we will analyze morphological changes in SAM size and primordia development under shade treatments. Confocal microscopy will be used to visualize fluorescent transcriptional reporters for stem cells (pCLV3:mCherry-NLS) and niche cells (pWUS:3xVENUS-NLS)³. This analysis will be complemented by studying phytochrome and key light signaling mutants, including PIFs and COP1. Since light effects on the meristem may be transient, we will conduct a detailed temporal analysis to identify the optimal time window for subsequent experiments. This objective will be led by the Swiss research group.

To map the spatial distribution of redox species in the SAM under controlled light conditions, we will combine tissue dissection with confocal microscopy and specific histochemical stain (DAB for H₂O₂, NBT for O₂^{•−}, DAF-FM for NO, and HBTP-H₂S for H₂S). These tools will help correlate redox gradients with SAM architecture and leaf development. Notably, these techniques offer sufficient temporal and spatial resolution to detect dynamic changes in the *Arabidopsis* shoot apex^{24,27,28}. To support this work, a PhD student from Argentina with experience in redox signal identification will undertake a 3-month visit to the Swiss lab, which provides essential expertise in shade signaling and SAM biology.

In the second phase, functional analyses will be conducted using specific donors and scavengers of each redox species. Additionally, *Arabidopsis* mutants impaired in the biosynthesis of these signals—such as *rbohcd, e, f* (O₂^{•−}/H₂O₂), *nia1nia2*, *noa1* (NO), and *des1, casc-1, lcd* (H₂S)—will be employed to determine their involvement in SAM regulation under shade. To fulfill this aim we will share genetic materials and chemicals among the involved labs. Two PhD students from the Latin American group will be partially dedicated to supporting this objective.

In the final phase, we will investigate if COP1 as a known redox-modified protein is involved in SAM regulation under shade by analyzing *cop1* mutants complemented with either wild-type or redox-insensitive (Cys-mutated) protein variants²⁴. In parallel, we will optimize alternative Biotin switch assays to detect redox PTMs induced by H₂O₂, NO, or H₂S specifically in the shoot apex enriched tissue^{26,31,32}. These assays, complemented with anti-biotin western blots, will reveal changes in the pattern of proteins modified by Redox PTMs under shade. At this stage, a Swiss researcher will visit the lab in Argentina to receive hands-on training in redox

switch assays and proteomic workflows. By the end of this project, we expect to generate have enough descriptive and functional data to prepare the project's first manuscript.

Task/Milestone (M)	Month											
	1	2	3	4	5	6	7	8	9	10	11	12
1-1. Spatio-temporal characterization of the developmental response to FR in the SAM												
1-2. Characterization of redox signals in response to FR in the SAM												
M1- Selection of the time frame and most relevant redox signal												
2. Evaluation of developmental responses to shade in redox-related mutants and under pharmacological treatments												
M2- Production of a working model of redox-dependent shade regulation of SAM development												
Preparation of first manuscript												
M3-Submission of first manuscript												
3.1. Evaluation of patterns of protein redox-PTM in response to FR												
3.2. Evaluation of developmental responses to shade in COP1 mutants in redox sensitive Cys												
M4- Optimization of protein redox-PTM in SAM enriched tissue for MS analysis												

Expected results and follow-up

This project will foster a new international collaboration between research groups led by two early-career female scientists based in Switzerland and Argentina. It will facilitate the exchange of personnel and expertise between teams with complementary areas of specialization, promoting knowledge transfer and capacity building across both institutions. This collaboration will be instrumental to set the basis to continue studying the role of redox signals on the developmental response to shade in the SAM. . In the subsequent step, mass spectrometry will be used to identify redox-sensitive target proteins that may integrate light and redox signaling in the SAM. The most promising candidates will be selected for functional characterization. We expect that this initial characterization will allow us to produce a first manuscript on the topic and will guide future functional studies and support applications for long-term collaborative funding:

-SNSF project <https://www.snf.ch/en/WAvYcY7awAUgoIST/funding/projects/projects-in-all-disciplines>

-SNSF Spirit <https://www.snf.ch/en/nlghrhzybd90TM9D/funding/programmes/spirit>

-HFSP <https://www.hfsp.org/funding/hfsp-funding/research-grants>

-ERC Synergy grant <https://erc.europa.eu/synergy-grant>

-ANPCYT PICT Raíces Grant <http://www.agencia.mincyt.gob.ar/frontend/agencia/instrumento/24>

- CONICET Postdoctoral fellowship <https://convocatorias.conicet.gov.ar/becas/convocatorias-vigentes/>

The outcomes of this project will provide key insights into how environmental signals, particularly light, are transduced into internal plant tissues to regulate development. By revealing the largely unexplored role of redox signals as central mediators between light cues and meristem activity, this research will uncover a fundamental regulatory layer in plant adaptation. These findings will advance our understanding of

developmental plasticity and lay the foundation for practical applications in sustainable agriculture. In the context of climate change, marked by rising temperatures and altered rainfall, an increment in crop density plantation is expected. Understanding shade responses becomes crucial for improving crop performance. The knowledge gained could inform breeding programs aimed at producing varieties with optimized architecture for dense planting or light stress. Additionally, identifying redox-regulated proteins may enable the development of biotechnological tools to directly modulate SAM activity for enhanced productivity.

Experiencie of the applicants

Applicant CH specializes in plant photomorphogenesis and thermomorphogenesis, with over 20 publications on phytochrome and phototropin signaling pathways. Her last research projects involve the study of the interaction between light signaling and tissue patterning, in leaves and hypocotyls. Her current interest is the regulation of meristematic activity by environmental cues such as shade and warm temperatures. She will contribute to the project with expertise on shade avoidance, tissue patterning, histochemical methods, confocal microscopy.

Applicant AR specializes in plant redox biology with around 25 publications in the field. Her research explores how environmental cues modulate redox signals, reprogramming plant development. Her group focuses on redox-based post-translational modifications, especially those involving cysteine residues, as key regulators of protein activity and signaling in light and temperature responses. The team brings strong expertise in redox signal detection, pharmacological approaches, and identification of redox-mediated PTMs—all essential to advancing the objectives of the proposed project.

Promotion of female and emerging researchers

All researchers involved in the project are women, and both principal investigators are within their first three years as independent group leaders. Collaborators are in their first or middle years of their PhD programs in their respective Universities in Argentina and Switzerland.

Intended Intellectual Property Rights

N/A

Ethical, safety, and regulatory issues

Arabidopsis transgenic work does not require ethical approval, but all biosafety protocols are followed at the respective institutions.

Bibliography

- 1 Fouracre, J. P. & Poethig, R. S. Lonely at the top? Regulation of shoot apical meristem activity by intrinsic and extrinsic factors. *Current opinion in plant biology* **58**, 17-24 (2020).
- 2 Yoshida, S., Mandel, T. & Kuhlemeier, C. Stem cell activation by light guides plant organogenesis. *Genes & Development* **25**, 1439-1450 (2011).
- 3 Pfeiffer, A. *et al.* Integration of light and metabolic signals for stem cell activation at the shoot apical meristem. *Elife* **5**, e17023 (2016).
- 4 Mohammed, B. *et al.* Converging light, energy and hormonal signaling control meristem activity, leaf initiation, and growth. **176**, 1365-1381 (2018).
- 5 Legris, M., Ince, Y. C. & Fankhauser, C. Molecular mechanisms underlying phytochrome-controlled morphogenesis in plants. *Nature communications* **10**, 5219 (2019). <https://doi.org/10.1038/s41467-019-13045-0>
- 6 Casal, J. J. & Fankhauser, C. Shade avoidance in the context of climate change. *Plant Physiology* **191**, 1475-1491 (2023). <https://doi.org/10.1093/plphys/kiad004>

- 7 Fernández-Milmanda, G. L. & Ballaré, C. L. Shade avoidance: expanding the color and hormone palette. *Trends in plant science* **26**, 509-523 (2021).
- 8 Kohnen, M. *et al.* Neighbor detection induces organ-specific transcriptomes, revealing patterns underlying hypocotyl-specific growth. *The Plant Cell*, tpc. 00463.02016 (2016).
- 9 Ballaré, C. L., Sánchez, R. A., Scopel, A. L., Casal, J. J. & Ghera, C. Early detection of neighbour plants by phytochrome perception of spectral changes in reflected sunlight. *Plant, Cell & Environment* **10**, 551-557 (1987).
- 10 Küpers, J. J. *et al.* Local light signaling at the leaf tip drives remote differential petiole growth through auxin-gibberellin dynamics. *Current Biology* **33**, 75-85. e75 (2023).
- 11 Legris, M. Light and temperature regulation of leaf morphogenesis in Arabidopsis. *New Phytologist* **240**, 2191-2196 (2023).
- 12 Nito, K. *et al.* Spatial regulation of the gene expression response to shade in Arabidopsis seedlings. *Plant and Cell Physiology* **56**, 1306-1319 (2015).
- 13 Carabelli, M. *et al.* Canopy shade causes a rapid and transient arrest in leaf development through auxin-induced cytokinin oxidase activity. *Genes & development* **21**, 1863-1868 (2007).
- 14 Hussain, E., Romanowski, A. & Halliday, K. J. PIF7 controls leaf cell proliferation through an AN3 substitution repression mechanism. *Proceedings of the National Academy of Sciences* **119**, e2115682119 (2022).
- 15 Romanowski, A., Furniss, J. J., Hussain, E. & Halliday, K. J. J. P. P. Phytochrome regulates cellular response plasticity and the basic molecular machinery of leaf development. (2021).
- 16 Chitwood, D. H. *et al.* Light-induced indeterminacy alters shade-avoiding tomato leaf morphology. *Plant physiology* **169**, 2030-2047 (2015).
- 17 Casson, S. A. & Hetherington, A. M. phytochrome B is required for light-mediated systemic control of stomatal development. *Current Biology* **24**, 1216-1221 (2014).
- 18 Pfeiffer, A., Wenzl, C. & Lohmann, J. U. Beyond flexibility: controlling stem cells in an ever changing environment. *Current opinion in plant biology* **35**, 117-123 (2017).
- 19 Mittler, R., Zandalinas, S. I., Fichman, Y. & Van Breusegem, F. Reactive oxygen species signalling in plant stress responses. *Nature reviews Molecular cell biology* **23**, 663-679 (2022).
- 20 Waszczak, C., Carmody, M. & Kangasjärvi, J. Reactive oxygen species in plant signaling. *Annual review of plant biology* **69**, 209-236 (2018).
- 21 Mishra, V., Singh, P., Tripathi, D. K., Corpas, F. J. & Singh, V. P. Nitric oxide and hydrogen sulfide: an indispensable combination for plant functioning. *Trends in Plant Science* **26**, 1270-1285 (2021).
- 22 Medrano-Macías, J. *et al.* Reactive oxygen, nitrogen, and sulfur species (RONSS) as a metabolic cluster for signaling and biostimulation of plants: an overview. *Plants* **11**, 3203 (2022).
- 23 Waszczak, C. *et al.* Oxidative post-translational modifications of cysteine residues in plant signal transduction. *Journal of experimental botany* **66**, 2923-2934 (2015).
- 24 Iglesias, M. J. *et al.* Shade-induced ROS/NO reinforce COP1-mediated diffuse cell growth. *Proceedings of the National Academy of Sciences* **121**, e2320187121 (2024).
- 25 Liu, Y. *et al.* Role of protein S-nitrosylation in plant growth and development. *Plant Cell Reports* **43**, 204 (2024).
- 26 Tian, Y. *et al.* Hydrogen peroxide positively regulates brassinosteroid signaling through oxidation of the BRASSINAZOLE-RESISTANT1 transcription factor. *Nature Communications* **9**, 1063 (2018).
- 27 Zeng, J. *et al.* Nitric oxide controls shoot meristem activity via regulation of DNA methylation. *Nature Communications* **14**, 8001 (2023).
- 28 Zeng, J., Dong, Z., Wu, H., Tian, Z. & Zhao, Z. Redox regulation of plant stem cell fate. *The EMBO journal* **36**, 2844-2855 (2017).
- 29 Wang, S., Liu, M., Hu, D., Dong, Z. & Zhao, Z. Control of DNA demethylation by superoxide anion in plant stem cells. *Nature Chemical Biology*, 1-10 (2024).
- 30 Zhang, T.-Q., Chen, Y. & Wang, J.-W. A single-cell analysis of the Arabidopsis vegetative shoot apex. *Developmental Cell* **56**, 1056-1074. e1058 (2021).

- 31 Jaffrey, S. R., Erdjument-Bromage, H., Ferris, C. D., Tempst, P. & Snyder, S. H. Protein S-nitrosylation: a physiological signal for neuronal nitric oxide. *Nature cell biology* **3**, 193-197 (2001).
- 32 Aroca, A., Jurado-Flores, A., Filipovic, M. R., Gotor, C. & Romero, L. C. in *Methods in Enzymology* Vol. 676 385-402 (Elsevier, 2022).